Prompt2Edit: Symbolic Planning for Vision–Language Image Editing

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***Abstract*—Recent prompt-based image editing methods rely on direct pixel-level manipulation through diffusion or enhancement models, which inherently limits fine-grained control over the sequencing of editing operations, target selection, and interpretability of the reasoning process. To address these limitations, we propose Prompt2Edit, a planning-centric framework that reframes image editing as a multimodal sequential decision-making problem. Given an input image and a natural-language instruction, Prompt2Edit first performs object-level scene parsing, then generates a structured sequence of symbolic editing actions—each specifying an operation type, a target object or region, and optional parameters—before delegating execution to external editing tools. The core planning module employs a lightweight Transformer architecture trained via imitation learning, operating on frozen vision–language embeddings and object representations. To circumvent the need for costly manual annotation, we introduce a synthetic data generation pipeline that leverages rule-based expert heuristics applied to object detections and templated instructions, yielding realistic, multi-step editing programs for scalable supervision. Experimental evaluation on a 62,500-sample synthetic dataset demonstrates high planning accuracy, achieving 86.3% action accuracy, 85.5% target-selection accuracy, and 83.6% full-sequence accuracy, improving over unimodal and component-removed variants. Qualitative results show that explicit planning yields coherent operation ordering and accurate spatial grounding, reducing common failure modes of one-shot editing approaches.**

***Index Terms*—Vision–language reasoning, Prompt-based image editing systems, Symbolic editing programs, Multimodal transformer planning, Object-centric visual analysis**

I. Introduction

PROMPT-BASED image editing has progressed rapidly with the development of diffusion models and vision–language encoders. Early systems such as InstructPix2Pix [1] and Stable Diffusion [2] advanced the capability to modify images directly from natural-language instructions, while multimodal encoders [3]–[5] enabled richer semantic grounding. Despite these improvements, most frameworks still treat editing as a single-step generative process, where the entire image is transformed at once based on a prompt [6]. While effective for global appearance changes, such approaches struggle when instructions involve multiple dependent operations, spatially localized modifications, or explicit ordering constraints, such as removing a foreground object before applying subsequent global adjustments.

In real-world scenarios, users frequently issue compound instructions such as "remove the person in the foreground, crop around the building, then brighten the image." Fulfilling such requests requires understanding not only what to edit but also in what sequence and on which visual entities. Existing tools typically lack explicit mechanisms for reasoning over editing intent, and as a result, they either require significant manual intervention or apply global transformations that ignore the compositional nature of the task [1], [7]. These limitations become more pronounced in situations where multiple objects, spatial relationships, or conflicting operations must be considered.

Meanwhile, recent work on vision–language models for robotics and navigation has explored structured planning formulations, in which high-level instructions are decomposed into sequences of interpretable actions [8]–[10]. These approaches show improved controllability and robustness in embodied decision-making tasks. However, similar planning-based formulations have been less explored in the context of image editing, where reasoning about how to perform edits may be as important as generating visually plausible outputs.

To fill this gap, we introduce the Prompt2Edit Planning Framework, which reformulates prompt-driven image editing as a symbolic planning problem. Instead of generating a final image directly from a prompt, the framework predicts an ordered sequence of symbolic editing operations, which are subsequently executed by task-specific editing tools. A lightweight transformer-based multimodal planner integrates object detections with language features to infer both the intended actions and their execution order. This design enables

fine-grained control, interpretability, and modularity, making it straightforward to substitute, extend, or optimize individual editing operations without retraining the planner.

Training such a planner normally requires significant human annotation, which is difficult and costly to obtain at scale. To address this, we propose a synthetic supervision strategy that automatically generates ground-truth editing programs using rule-based logic applied to YOLO-derived [11] object layouts. This approach, inspired by advances in vision–language reasoning and program synthesis [12], allows the model to learn complex editing behaviors without curated datasets.

Our contributions are summarized as follows:

- We cast image editing as a structured planning problem that predicts ordered, interpretable action sequences instead of performing a single generative transformation.
- We introduce structure-aware regularization to enforce semantic validity and consistency.
- We introduce a rule-driven data generation strategy that enables scalable training of editing plans without human annotation.

II. Methodology

Prompt2edit planning formulates natural-language image editing as a structured, multi-step decision problem. Given an input image and a user instruction, the goal is to predict an ordered sequence of symbolic editing operations that can be executed by external image editing tools. Unlike end-to-end generative editing approaches [13], [14], the proposed method explicitly separates planning from execution, enabling interpretability, controllability, and modular extensibility.

An overview of the framework is illustrated in Fig. 1. The system consists of three components: object-level image analysis, multimodal edit planning, and tool-based execution. Only the planning module is trained; all perceptual encoders and executors are fixed.

A. Problem Definition

Let I denote an input image and P a natural-language editing instruction. The objective is to generate a fixed-length editing program

$$\mathcal{E} = \{e_1, e_2, \ldots, e_K\}, \quad K = 3, \tag{1}$$

where each step e_k is a tuple

$$e_k = (a_k, o_k, \pi_k). \tag{2}$$

Here, $a_k \in \mathcal{A}$ is the editing action selected from a predefined action set $\mathcal{A} = \{\textit{crop}, \textit{enhance}, \textit{diffusion}, \textit{noop}\}$. The variable o_k denotes the target of the action, defined as either an object index detected in the image or a special global target. The parameter π_k is action-dependent: it represents an integer enhancement strength for the enhance action, a natural-language tool prompt for the diffusion action, and is undefined for crop and noop. The program $\mathcal{E}$ specifies what operations to perform and where, while the execution module determines how these operations are applied at the pixel level.

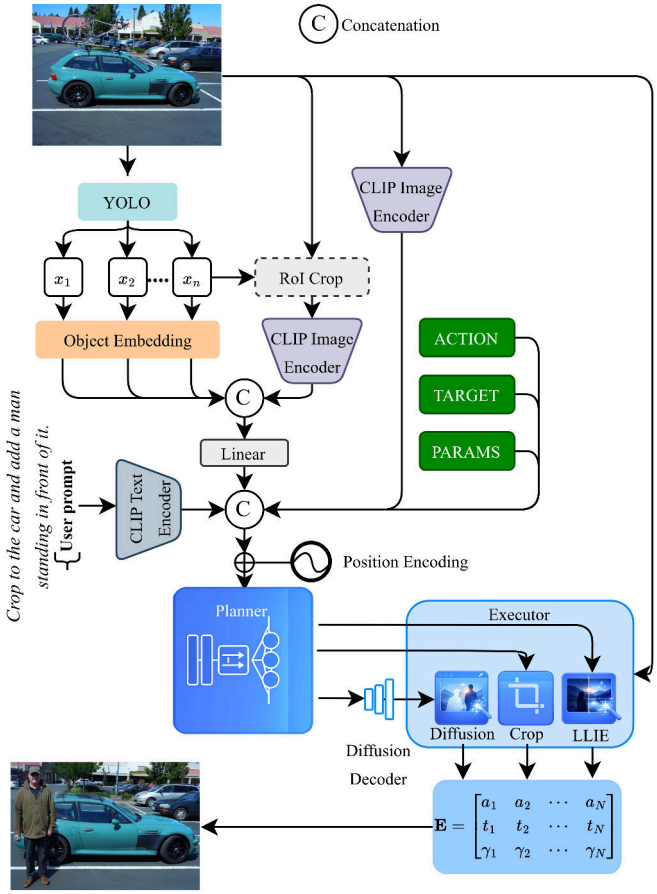

Fig. 1. Overview of the Prompt2Edit Planning Framework.

The formulation follows symbolic decision prediction frameworks used in program induction and vision–language planning [15].

B. Object-Level Image Representation

To ground editing decisions in visual content, object-level representations are extracted from the input image using a frozen object detector [11]. Given image I, the detector produces a set of N objects with class labels and bounding boxes. Each object i is encoded as

$$x_i = [\phi(c_i); b_i; v_i], \tag{3}$$

here c_i is the object class, b_i its bounding box coordinates, and v_i a visual embedding extracted using a frozen CLIP image encoder [4] backbone. The object embeddings $\{x_i\}_{i=1}^{N}$ form a structured representation of the scene and define the candidate target space for planning.

C. Instruction Encoding

The user instruction P is embedded using a pretrained CLIP language encoder [4], which remains frozen during training. The resulting text embedding $t = f_{\text{text}}(P)$ captures the semantic intent of the instruction [16] but does not directly specify executable actions. Instead, it serves as input to the

planner, which learns to map language semantics to symbolic editing decisions.

D. Multimodal Planning Network

The core of the framework is a lightweight transformer-based planner that reasons jointly over the instruction embedding t and the set of object embeddings $\{x_i\}$, similar in spirit to neural program generators and vision-language planners [17], [18]. We concatenate the instruction token with object tokens and apply self-attention over the combined set. We introduce K learnable step tokens, which attend to the instruction and object tokens. The contextualized outputs of these step tokens are used as $\{h_k\}_{k=1}^{K}$, one for each planning step. From each hidden state h_k, separate prediction heads output the components of e_k.

The action head produces a probability distribution $\hat{p}_k(a)$ over the action set $\mathcal{A}$. The target head outputs a distribution $\hat{p}_k(o)$ over all detected objects and a global target. For enhance actions, an additional head predicts an integer enhancement strength. For diffusion actions, a conditional text decoder generates a sequence of tokens forming an optimized tool prompt to be consumed by the diffusion executor. All planner parameters are learned from scratch, while perceptual encoders remain frozen.

E. Tool-Based Execution

The predicted program $\mathcal{E}$ is executed by external, non-trainable tools. Cropping produces the final output by selecting the target bounding box. Enhancement actions operate on the target region using a fixed enhancement model, with region extraction and reintegration handled by helper functions. Diffusion actions are executed by a pretrained diffusion model (InstructPix2Pix [1]) using the generated tool prompt and a binary mask derived from the target bounding box. For diffusion actions, the prompt parameter is generated by a lightweight language decoder initialized from a pretrained GPT-2 model [19] and fine-tuned via supervised imitation learning on teacher-generated tool prompts [20]. The diffusion executor itself remains frozen.

The execution stage is entirely decoupled from training and is not involved in gradient computation. This decoupling prevents gradient propagation through image generators and allows heterogeneous tools to be integrated without retraining the planner, consistent with recent tool-augmented reasoning and acting frameworks [21], [22]. The design and implementation of execution tools, including mask construction and pixel-level manipulation, are outside the scope of this work; our focus is exclusively on learning the symbolic planning policy.

F. Training Objective

The planner is trained using supervised imitation learning from synthetically generated expert programs. For each training sample, ground-truth labels (a_k^*, o_k^*, π_k^*) are available for all steps.

The action prediction is supervised using cross-entropy:

$$\mathcal{L}_{act} = \sum_{k=1}^{K} CE(\hat{p}_k(a), a_k^*). \tag{4}$$

For all non-noop steps, the target prediction is trained using cross-entropy:

$$\mathcal{L}_{tgt} = \sum_{k=1}^{K} \mathbf{1}[a_k^* \neq noop] CE(\hat{p}_k(o), o_k^*). \tag{5}$$

For enhance actions, the enhancement strength is supervised as a discrete classification problem:

$$\mathcal{L}_{enh} = \sum_{k=1}^{K} \mathbf{1}[a_k^* = enhance] CE(\hat{p}_k(s), s_k^*), \tag{6}$$

where s_k^* denotes the ground-truth enhancement level.

For diffusion actions, the planner generates a sequence of token distributions $\hat{p}_{k,t}(y)$ over a vocabulary of size V. Given the ground-truth token sequence $\{y_{k,t}^*\}$, the diffusion loss is defined as

$$\mathcal{L}_{diff} = \sum_{k=1}^{K} \mathbf{1}[a_k^* = diffusion] \sum_{t=1}^{L} CE(\hat{p}_{k,t}(y), y_{k,t}^*), \tag{7}$$

where padding tokens are ignored.

Following prior work on constraint-aware program learning and symbolic planning [23], we incorporate auxiliary structure-aware regularization terms that encode domain-consistent planning priors. A consistency regularizer penalizes abrupt changes in target selection across logically related steps:

$$\mathcal{L}_{cons} = \sum_{k=2}^{K} \mathbf{1}[(a_{k-1}^*, a_k^*) \in \mathcal{R}] \left(1 - \hat{p}_{k-1}(o)^\top \hat{p}_k(o)\right), \tag{8}$$

where $\mathcal{R}$ denotes a predefined set of action pairs for which target consistency is expected.

To discourage redundant consecutive operations, a smoothness penalty is introduced:

$$\begin{aligned} \mathcal{L}_{smooth} = \sum_{k=2}^{K} & \mathbf{1}\left[a_{k-1}^* \neq noop \wedge a_k^* \neq noop\right] \\ & \cdot \left(\hat{p}_{k-1}(a)^\top \hat{p}_k(a)\right) \left(\hat{p}_{k-1}(o)^\top \hat{p}_k(o)\right). \end{aligned} \tag{9}$$

The final training objective is

$$\begin{aligned} \mathcal{L} = & \lambda_{act}\mathcal{L}_{act} + \lambda_{tgt}\mathcal{L}_{tgt} + \lambda_{enh}\mathcal{L}_{enh} \\ & + \lambda_{diff}\mathcal{L}_{diff} + \lambda_{cons}\mathcal{L}_{cons} + \lambda_{smooth}\mathcal{L}_{smooth}. \end{aligned} \tag{10}$$

III. Experiments and Results

This section evaluates the proposed Prompt2Edit framework with respect to its ability to generate correct symbolic editing plans and to execute them effectively using a diffusion-based image editing model. Both quantitative planning accuracy and qualitative visual outcomes are analyzed to assess the contribution of explicit planning under different editing scenarios.

TABLE I
DATASET STATISTICS CONSTRUCTED FROM MS-COCO

Split	Images	Samples	Avg. Samples
Training	10,000	52,300	5.23
Validation	1,000	5,120	5.12
Test	1,000	5,080	5.12

TABLE II
PLANNING ACCURACY (%).THE BEST RESULTS ARE SHOWN IN **BOLD**.

Method	Action	Target	Full Sequence
w/o Obj	57.1	32.3	26.7
w/o Global	48.9	67	44
w/o Lang	12.8	43.1	10.8
Ours	**86.3**	**85.5**	**83.6**

TABLE III
PROGRAM-LEVEL EVALUATION OF SYMBOLIC EDITING PLANS USING OOA, RSP, AND AOC METRICS. THE BEST RESULTS ARE SHOWN IN **BOLD**.

Metric	Ours	w/o Obj	w/o Global	w/o Lang
OOA	**88.1**	52.1	36.3	14
RSP	**84.2**	24	61	19.8
AOC	**87.6**	48.1	68.8	23.5

A. Experimental Setup

All experiments are conducted on a single NVIDIA RTX 4090 GPU. The planner is implemented as a lightweight Transformer with two layers, a hidden dimension of 256, and four attention heads. We fix the program length to $K = 3$ to align with the three available executors. Instructions requiring fewer operations are padded with NOOP actions, while instructions requiring more than three distinct operations fall outside the scope of this introductory study.

The model is trained using the loss function defined in Section II-F with loss weights set to $\lambda_{act} = 1.0$, $\lambda_{tgt} = 1.0$, $\lambda_{enh} = 0.5$, $\lambda_{dif} = 1.0$, $\lambda_{cons} = 0.3$, and $\lambda_{smooth} = 0.2$. Optimization is performed using Adam with a learning rate of 1×10^{-4} and a batch size of 32. Training converges after approximately 100 epochs.

Due to the lack of prior methods that explicitly generate executable symbolic programs for image editing, direct quantitative comparison is not possible. Therefore, we evaluate our approach through extensive component comparisons that remove or modify key components of the planner, which serves as a proxy for baseline comparison.

B. Dataset

As no pubic dataset exists for prompt-based image editing with symbolic planning supervision, we construct a synthetic dataset from MS-COCO 2017 [24] images. A subset of images is randomly sampled and split at the image level into training, validation, and test sets. For each image, multiple natural-language editing instructions are generated automatically based on detected objects and their spatial relations. Ground-truth symbolic editing programs are produced using rule-based heuristics for crop and enhance actions, while diffusion parameters are generated by an offline teacher vision-language model [5], which produces optimized text prompts conditioned on the image and instruction. Each training sample consists of an image, an instruction, and a three-step symbolic program. On average, each image yields approximately five samples. Dataset statistics are reported in Table I.

C. Quantitative and Qualitative Results

Table II reports step-level planning accuracy. The full model achieves high action, target, and full-sequence accuracy. Removing object tokens mainly harms target selection, removing the global token disrupts step ordering, and removing language severely degrades all metrics.

Table III presents program-level evaluation metrics that characterize the structural quality of predicted editing plans. Operation Ordering Accuracy (OOA) quantifies whether actions are predicted in a semantically valid sequence, Region/Scope Precision (RSP) measures spatial correctness, and Action-Object Consistency (AOC) evaluates alignment between predicted operations and selected targets. The proposed model consistently outperforms all variants across these metrics, demonstrating that joint modeling of language, objects, and global context is critical for coherent plan synthesis. In contrast, component-removed variants exhibit complementary failure modes, such as incorrect ordering, scope leakage, or inconsistent action–target pairing. The full model maintains high consistency across all dimensions, while component-removed variants often apply operations to incorrect regions or produce semantically invalid sequences.

Qualitative results further clarify how explicit planning affects both operation ordering and downstream execution quality. Fig. 2 reports qualitative results for the instruction "remove the person and improve the luminosity". Although the prompt lists object removal before enhancement, directly applying removal first can introduce inpainting artifacts that are subsequently amplified by global brightness adjustment. In such cases, the planner infers an execution order that reduces error propagation while remaining consistent with the instruction intent. Specifically, it applies low-light enhancement using the adopted operator [25] to improve visibility and stabilize the image, followed by object removal, resulting in more reliable editing outcomes. In contrast, direct prompt-based editing [1] fails to preserve structural consistency. The variant w/o Obj mislocalizes the target and removes unintended regions. The model w/o Global predicts an incorrect operation order, leading to amplified artifacts after enhancement. The language-agnostic variant w/o Lang ignores the removal instruction entirely and applies only enhancement. The full Prompt2Edit model correctly infers both the operation order and enhancement strength, yielding a clean and coherent result.

To further evaluate the planner's ability to handle global instructions, we conduct an additional qualitative test in which the prompt requires modifying the sky rather than manipu-

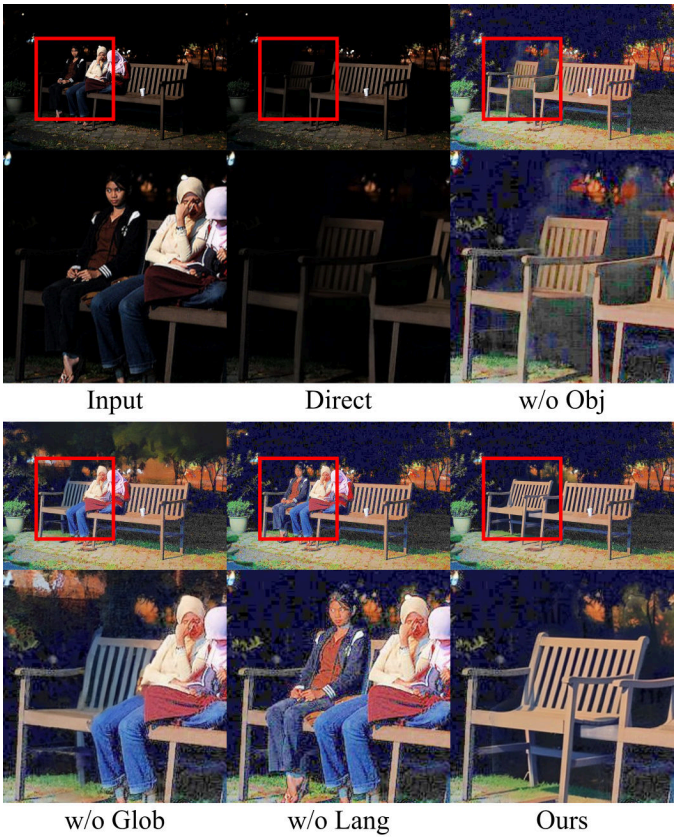

Fig. 2. Qualitative results for the prompt "remove the person and improve the luminosity". The component-removed variants fail in complementary ways, while our full model produces the correct multi-step edit.

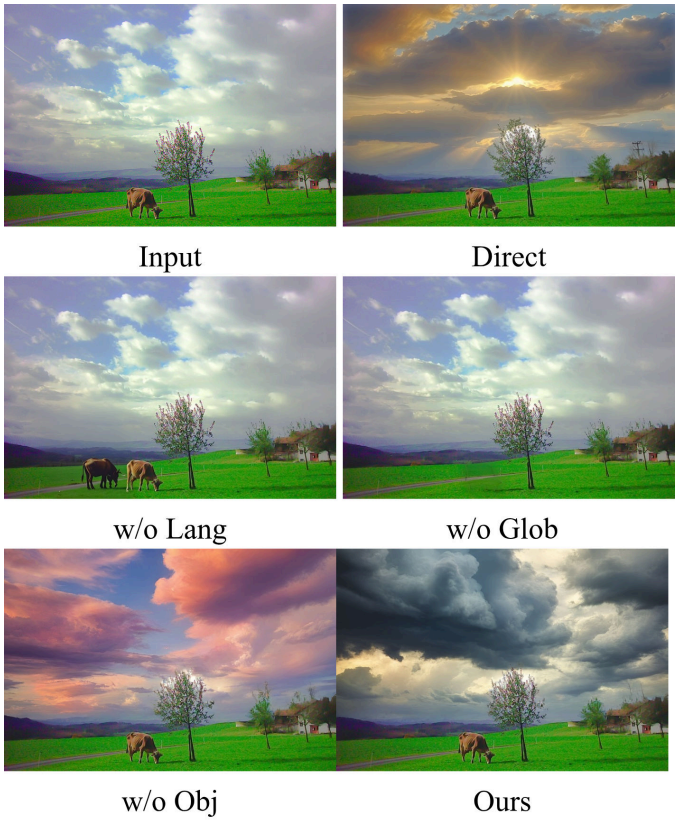

Fig. 3. Program-level planning evaluation across different model variants.

lating a specific object. As shown in Fig. 3, this scenario reveals how each component contributes to maintaining scene consistency even when the editing operation is conceptually simple.

Direct editing produces a visually coherent enhancement because the instruction correspond to an end-to-end image editing, requiring no step-specific reasoning. However, the component-removed variants demonstrate that planning cues remain essential for avoiding unintended structural changes. The w/o Lang model introduces an additional animal in the foreground, indicating that without semantic grounding the planner cannot restrict the modification to the sky. The w/o Global model misinterprets the scene layout and removes the horse entirely, leaving an artifact where the animal previously stood; this confirms that global contextual information is necessary for distinguishing foreground entities from background regions. The w/o Obj model succeeds in altering the sky but fails to ensure clean separation between global edits and object boundaries, revealing a lack of structural awareness.

In contrast, the full Prompt2Edit framework accurately confines the transformation to the sky and preserves all foreground elements without hallucinations or deletions. This result demonstrates that the proposed symbolic planning approach provides robustness and spatial discipline even for visually broad editing tasks.

IV. Conclusions

We introduced Prompt2Edit, a planning-centered framework that translates natural-language instructions into structured, symbolic editing programs grounded in object-level visual reasoning. By decoupling high-level decision processes from low-level manipulation, the framework enables a form of image editing that is both interpretable and controllable-qualities that are often lacking in diffusion-based or end-to-end generative approaches. Our results demonstrate that even a lightweight multimodal transformer, when equipped with reliable object cues and global contextual information, can learn to predict coherent multi-step procedures that align closely with user intent. Beyond measurable improvements in action selection, region grounding, and program consistency, the framework highlights the broader value of reasoning-first workflows in visual editing. Rather than entangling planning with execution, Prompt2Edit treats editing as a symbolic decision-making problem. This design choice not only simplifies debugging and error attribution but also allows the editing system to adapt to different execution modules without retraining. Future work may integrate segmentation-level grounding, richer spatial descriptors, and more diverse editing operators to expand the expressiveness of the predicted programs. Learning from human demonstrations or from generative feedback could further refine the planner's ability to handle nuanced or abstract

prompts. Finally, extending the planning paradigm to video or interactive editing interfaces may provide a unified framework for controllable, temporally consistent editing across modalities.

ACKNOWLEDGMENT

This research was supported by the Science and Technology Project of Sichuan (Grant No. 2024ZHCG0192). The authors also thank Science Hope for providing access to their editing tools used for evaluation and deployment testing.

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